

1. List up to 10 different protocols that appear in the Protocol column in the unfiltered packet-listing window.



Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.31.226.96	8.8.8.8	ICMP	58	Echo (ping) request id=0x0001, seq=6866/53786, ttl=255 (reply in 2)
2	0.167725000	8.8.8.8	10.31.226.96	ICMP	58	Echo (ping) reply id=0x0001, seq=6866/53786, ttl=114 (request in 1)
3	0.671188300	d2:6c:d5:22:00:3a	AzureWaveTec_2e:a8:54	ARP	42	Who has 10.31.226.96? Tell 10.31.226.85
4	0.671219000	AzureWaveTec_2e:a8:54	d2:6c:d5:22:00:3a	ARP	42	10.31.226.96 is at 9c:c7:d3:2e:a8:54
5	0.737984000	10.31.226.96	172.188.155.25	TCP	55	57319 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
6	0.866507400	172.188.155.25	10.31.226.96	TCP	66	443 → 57319 [ACK] Seq=1 Ack=2 Win=307 Len=0 SLE=1 SRE=2
7	1.168841000	10.31.226.96	8.8.8.8	ICMP	58	Echo (ping) request id=0x0001, seq=6867/54042, ttl=255 (reply in 8)
8	1.219514700	8.8.8.8	10.31.226.96	ICMP	58	Echo (ping) reply id=0x0001, seq=6867/54042, ttl=114 (request in 7)
9	1.256253700	10.31.226.96	10.31.226.85	DNS	89	Standard query 0x65b8 HTTPS clientservices.googleapis.com
10	1.256523200	10.31.226.96	10.31.226.85	DNS	89	Standard query 0x4ea8 AAAA clientservices.googleapis.com
11	1.256677900	10.31.226.96	10.31.226.85	DNS	89	Standard query 0x22d0 A clientservices.googleapis.com
12	1.283172800	10.31.226.96	10.31.226.85	DNS	94	Standard query 0x5ead HTTPS oauthaccountmanager.googleapis.com
13	1.283547000	10.31.226.96	10.31.226.85	DNS	94	Standard query 0xd9f2 AAAA oauthaccountmanager.googleapis.com
14	1.283716100	10.31.226.96	10.31.226.85	DNS	94	Standard query 0x4611 A oauthaccountmanager.googleapis.com
15	1.284949700	10.31.226.96	10.31.226.85	DNS	79	Standard query 0xd29 HTTPS accounts.google.com
16	1.285345700	10.31.226.96	10.31.226.85	DNS	79	Standard query 0xde3f AAAA accounts.google.com
17	1.285496000	10.31.226.96	10.31.226.85	DNS	79	Standard query 0xb517 A accounts.google.com
18	1.286099300	10.31.226.96	10.31.226.85	DNS	74	Standard query 0x1741 HTTPS www.google.com
19	1.286228700	10.31.226.96	10.31.226.85	DNS	74	Standard query 0xa617 AAAA www.google.com
20	1.286339200	10.31.226.96	10.31.226.85	DNS	74	Standard query 0xe45d A www.google.com
21	1.319145600	10.31.226.85	10.31.226.96	DNS	201	Standard query response 0x4ea8 AAAA clientservices.googleapis.com AAAA 2404:6800:4000:1025::54
22	1.319147700	10.31.226.85	10.31.226.96	DNS	146	Standard query response 0x65b8 HTTPS clientservices.googleapis.com SOA ns1.google.com
23	1.320517900	10.31.226.85	10.31.226.96	DNS	185	Standard query response 0x22d0 A clientservices.googleapis.com A 192.178.173.101 A 192.178.173.101
24	1.326625300	2401:4900:ac10:5375:f5f8:bd83:e5f:c8cb	2404:6800:4000:1010::64	TCP	86	60930 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM=1
25	1.383263800	10.31.226.85	10.31.226.96	DNS	318	Standard query response 0xd9f2 AAAA oauthaccountmanager.googleapis.com AAAA 2001:4860:4827:7700::
26	1.383265500	10.31.226.85	10.31.226.96	DNS	129	Standard query response 0xd29 HTTPS accounts.google.com SOA ns1.google.com
27	1.383266000	10.31.226.85	10.31.226.96	DNS	95	Standard query response 0xb517 A accounts.google.com A 192.178.211.84
28	1.383266200	10.31.226.85	10.31.226.96	DNS	107	Standard query response 0xde3f AAAA accounts.google.com AAAA 2404:6800:4000:1025::54
29	1.383266600	10.31.226.85	10.31.226.96	DNS	151	Standard query response 0x5ead HTTPS oauthaccountmanager.googleapis.com SOA ns1.google.com
30	1.383267000	10.31.226.85	10.31.226.96	DNS	202	Standard query response 0xe45d A www.google.com A 142.251.150.119 A 142.251.156.119 A
31	1.383267200	10.31.226.85	10.31.226.96	DNS	99	Standard query response 0x1741 HTTPS www.google.com HTTPS
32	1.383268500	10.31.226.85	10.31.226.96	DNS	298	Standard query response 0xa617 AAAA www.google.com AAAA 2001:4860:4827:7700:: AAAA 2001:4860:4827:7700::
33	1.383269100	10.31.226.85	10.31.226.96	DNS	222	Standard query response 0x4611 A oauthaccountmanager.googleapis.com A 172.217.115.4 A 172.217.115.4
34	1.390227000	10.31.226.96	10.31.226.85	DNS	81	Standard query 0x7227 HTTPS update.googleapis.com
35	1.390526400	10.31.226.96	10.31.226.85	DNS	81	Standard query 0x23fe AAAA update.googleapis.com
36	1.390680500	10.31.226.96	10.31.226.85	DNS	81	Standard query 0x949a A update.googleapis.com
37	1.390737500	2401:4900:ac10:5375:f5f8:bd83:e5f:c8cb	2001:4860:4827:7700::	QUIC	1292	Initial, DCID=f6d8e2cdea3e662b, PKN: 1, CRYPTO, CRYPTO, PING, CRYPTO, CRYPTO, PING, PING
38	1.390763100	2401:4900:ac10:5375:f5f8:bd83:e5f:c8cb	2001:4860:4827:7700::	QUIC	1292	Initial, DCID=f6d8e2cdea3e662b, PKN: 2, CRYPTO, PING, PING, CRYPTO, PADDING, CRYPTO, PING
39	1.390773300	2401:4900:ac10:5375:f5f8:bd83:e5f:c8cb	2404:6800:4000:1025::54	TCP	86	60930 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM=1
40	1.390827700	2401:4900:ac10:5375:f5f8:bd83:e5f:c8cb	2001:4860:4827:7700::	TCP	86	60930 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM=1

0000 d2 6c d5 22 00 3a 9c c7 d3 2e a8 54 08 00 45 00 1 : * : : : T : E :

Packets: 15875 - Dropped: 0 (0.0%) Profile: Default

2. What is the Internet address of the www.gujaratuniversity.ac.in?

What is the Internet address of your computer?

->nslookup www.gujaratuniversity.ac.in

->ipconfig



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JAY PATEL>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::fb58:6deb:3ef4:735a%12
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Wi-Fi 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi 5:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

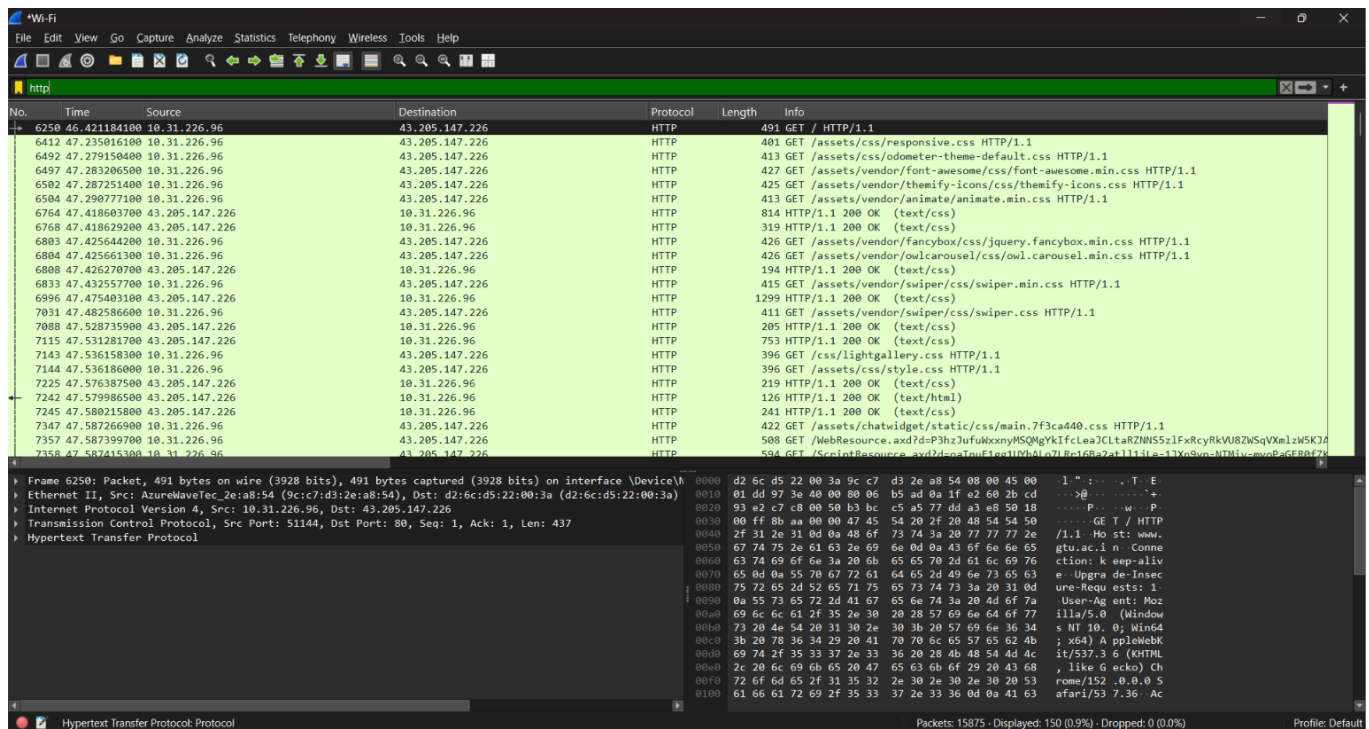
    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:ac10:5375:fd3d:99cd:38f5:ec63
    Temporary IPv6 Address. . . . . : 2401:4900:ac10:5375:f5f8:bd83:e5f:c8cb
    IPv6 Address. . . . . : 2401:4900:ac1f:c9b0:3ed5:f7bb:903a:6ce8
    Temporary IPv6 Address. . . . . : 2401:4900:ac1f:c9b0:e9ea:4829:da9d:27bb
    Link-local IPv6 Address . . . . . : fe80::a7ef:8b5d:4d62:4bdc%13
    IPv4 Address. . . . . : 10.31.226.96
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::d06c:d5ff:fe22:3a%13
                                10.31.226.85

Ethernet adapter Bluetooth Network Connection:
```

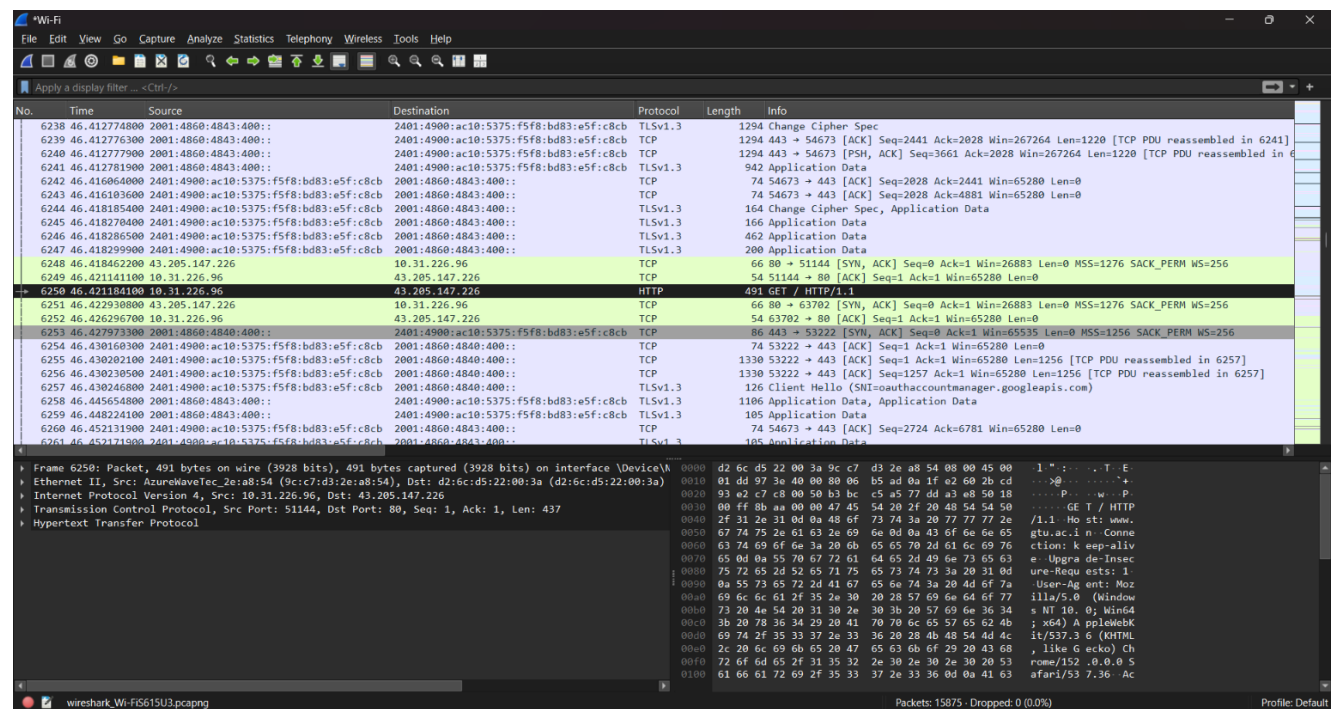
```
C:\Users\JAY PATEL>nslookup www.gtu.ac.in
Server: UnKnown
Address: 10.31.226.85

Non-authoritative answer:
Name: www.gtu.ac.in
Addresses: 43.205.147.226
           13.235.221.101
```

➔ Total HTTP : 150



➔Total Pcket : 15875



5. Now, use display filters to determine how many packets contain your ip address (hint: Use ip.addr instead of the clumsy ip.src or ip.dst format).



ip.addr==10.31.226.85

Ip address = 448 packets

The screenshot shows the Wireshark network protocol analyzer interface. The display filter is set to `ip.addr==10.31.226.85`. The packet list pane shows 448 packets, all of which are DNS queries and responses to 10.31.226.85. The packet details pane shows the structure of a DNS query packet, including the Ethernet II header, Internet Protocol Version 4 header, User Datagram Protocol header, and Domain Name System (query) header.

No.	Time	Source	Destination	Protocol	Length	Info
9	1.256253700	10.31.226.96	10.31.226.85	DNS	89	Standard query 0x65b8 HTTPS clientservices.googleapis.com
10	1.256523200	10.31.226.96	10.31.226.85	DNS	89	Standard query 0x4ea8 AAAA clientservices.googleapis.com
11	1.256677900	10.31.226.96	10.31.226.85	DNS	89	Standard query 0x22d0 A clientservices.googleapis.com
12	1.283172800	10.31.226.96	10.31.226.85	DNS	94	Standard query 0x5ead HTTPS oauthaccountmanager.googleapis.com
13	1.283547000	10.31.226.96	10.31.226.85	DNS	94	Standard query 0xd9f2 AAAA oauthaccountmanager.googleapis.com
14	1.283716100	10.31.226.96	10.31.226.85	DNS	94	Standard query 0x4611 A oauthaccountmanager.googleapis.com
15	1.284949700	10.31.226.96	10.31.226.85	DNS	79	Standard query 0xde29 HTTPS accounts.google.com
16	1.285345700	10.31.226.96	10.31.226.85	DNS	79	Standard query 0xde3f AAAA accounts.google.com
17	1.285496000	10.31.226.96	10.31.226.85	DNS	79	Standard query 0xb517 A accounts.google.com
18	1.286099300	10.31.226.96	10.31.226.85	DNS	74	Standard query 0x1741 HTTPS www.google.com
19	1.286228700	10.31.226.96	10.31.226.85	DNS	74	Standard query 0xa617 AAAA www.google.com
20	1.286339200	10.31.226.96	10.31.226.85	DNS	74	Standard query 0xe45d A www.google.com
21	1.319145600	10.31.226.85	10.31.226.96	DNS	201	Standard query response 0x4ea8 AAAA clientservices.googleapis.com AAAA 2404:6800:4000:...
22	1.319147700	10.31.226.85	10.31.226.96	DNS	146	Standard query response 0x65b8 HTTPS clientservices.googleapis.com SOA ns1.google.com
23	1.320517900	10.31.226.85	10.31.226.96	DNS	185	Standard query response 0x22d0 A clientservices.googleapis.com A 192.178.173.101 A 192...
25	1.383263800	10.31.226.85	10.31.226.96	DNS	318	Standard query response 0xd9f2 AAAA oauthaccountmanager.googleapis.com AAAA 2001:4860:...
26	1.383265500	10.31.226.85	10.31.226.96	DNS	129	Standard query response 0xde29 HTTPS accounts.google.com SOA ns1.google.com
27	1.383266000	10.31.226.85	10.31.226.96	DNS	95	Standard query response 0xb517 A accounts.google.com A 192.178.211.84
28	1.383266200	10.31.226.85	10.31.226.96	DNS	107	Standard query response 0xde3f AAAA accounts.google.com AAAA 2404:6800:4000:1025:54
29	1.383266600	10.31.226.85	10.31.226.96	DNS	151	Standard query response 0x5ead HTTPS oauthaccountmanager.googleapis.com SOA ns1.google...
30	1.383267000	10.31.226.85	10.31.226.96	DNS	202	Standard query response 0xe45d A www.google.com A 142.251.150.119 A 142.251.156.119 A
31	1.383267200	10.31.226.85	10.31.226.96	DNS	99	Standard query response 0x1741 HTTPS www.google.com HTTPS
32	1.383268500	10.31.226.85	10.31.226.96	DNS	298	Standard query response 0xa617 AAAA www.google.com AAAA 2001:4860:4827:7700: AAAA 200...
33	1.383269100	10.31.226.85	10.31.226.96	DNS	222	Standard query response 0x4611 A oauthaccountmanager.googleapis.com A 172.217.115.4 A

Frame 9: Packet, 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on interface \Device\NPF_{7C4...}

Ethernet II, Src: AzureWaveTec_2e:a8:54 (9c:c7:d3:2e:a8:54), Dst: d2:6c:d5:22:00:3a (d2:6c:d5:22:00:3a)

Internet Protocol Version 4, Src: 10.31.226.96, Dst: 10.31.226.85

User Datagram Protocol, Src Port: 62207, Dst Port: 53

Domain Name System (query)

0000 d2 6c d5 22 00 3a 9c c7 d3 2e a8 54 08 00 45 00 .1 T . E .

0010 00 4b 1a fc 00 00 00 11 46 b2 0a 1f e2 60 0a 1f .K F

0020 e2 55 f2 ff 00 35 00 37 02 b3 65 b8 01 00 00 01 .U . 5 7 . a

0030 00 00 00 00 00 00 0e 63 6c 69 65 6e 74 73 65 72 c l i e n t s e r

0040 76 69 63 65 73 0a 67 6f 6f 67 6c 65 61 70 69 73 . v i c e s . g o o g l e a p i s

0050 03 63 6f 6d 00 00 41 00 01 . c o m . A

Wireshark - Wi-Fi-HS61SU3.pcapng

Packets: 15875 - Displayed: 448 (2.8%) - Dropped: 0 (0.0%)

Profile: Default

6. What is this filter you used? Now, reverse the filter to determine how many packets don't contain your ip address.

➔ `!ip.addr==10.31.226.85`

Ip address: 15427 packets

The screenshot shows the Wireshark network protocol analyzer. The filter bar at the top contains the filter `!ip.addr==10.31.226.85`. The packet list pane displays a list of captured packets, including ICMP Echo (ping) requests and replies, ARP requests, and TCP SYN and ACK packets. The packet details pane for packet 24 shows the structure of the packet: Ethernet II (Source: AzureWaveTec_2e:a8:54, Destination: d2:6c:d5:22:00:3a), Internet Protocol Version 6 (Source: 2401:4900:ac10:5375:f5f8:bd83:e5f:c8cb, Destination: 2404:6800:4000:1025::54), and Transmission Control Protocol (Source Port: 56733, Destination Port: 443, Sequence Number: 0, Length: 0). The packet bytes pane shows the raw data in hexadecimal and ASCII.

7. What is the IP address of your computer? Of the gtu.ac.in server?



```
Command Prompt

Wireless LAN adapter Wi-Fi 5:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:
    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2401:4900:ac10:5375:fd3d:99cd:38f5:ec63
    Temporary IPv6 Address. . . . . : 2401:4900:ac10:5375:f5f8:bd83:e5f:c8cb
    IPv6 Address. . . . . : 2401:4900:ac1f:c9b0:3ed5:f7bb:903a:6ce8
    Temporary IPv6 Address. . . . . : 2401:4900:ac1f:c9b0:e9ea:4829:da9d:27bb
    Link-Local IPv6 Address . . . . . : fe80::a7ef:8b5d:4d62:4bdc%13
    IPv4 Address. . . . . : 10.31.226.96
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::d06c:d5ff:fe22:3a%13
                                10.31.226.85

Ethernet adapter Bluetooth Network Connection:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\JAY PATEL>nslookup www.gtu.ac.in
Server:      Unknown
Address:     10.31.226.85

Non-authoritative answer:
Name:       www.gtu.ac.in
Addresses:  43.205.147.226
            13.235.221.101
```

8. What HTTP version is your browser running? What version of HTTP is the server running?

➔ Request Version: HTTP/1.1

The image shows a Wireshark packet capture of an HTTP request. The packet list on the left shows a GET request for `/assets/vendor/fancybox/css/jquery.fancybox.min.css` at time 6803.47. The packet details pane on the right shows the request structure:

- Frame 6803: Packet, 426 bytes on wire (3408 bits), 426 bytes captured (3408 bits) on interface DeviceN
- Ethernet II, Src: AzureWaveTec_2e:a8:54 (9c:c7:d3:2e:a8:54), Dst: d2:6c:d5:22:00:3a (d2:6c:d5:22:00:3a)
- Internet Protocol Version 4, Src: 10.31.226.96, Dst: 43.205.147.226
- Transmission Control Protocol, Src Port: 63702, Dst Port: 80, Seq: 348, Ack: 5861, Len: 372
- Hypertext Transfer Protocol
 - GET /assets/vendor/fancybox/css/jquery.fancybox.min.css HTTP/1.1
 - Request Method: GET
 - Request URI: /assets/vendor/fancybox/css/jquery.fancybox.min.css
 - Request Version: HTTP/1.1
 - Host: www.gtu.ac.in
 - Connection: keep-alive
 - User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/100.0.0.0 Safari/537.36
 - Accept: text/css,*/*;q=0.1
 - Referer: http://www.gtu.ac.in
 - Accept-Encoding: gzip, deflate
 - Accept-Language: en-US,en;q=0.9,gu;q=0.8

The packet bytes pane on the right shows the raw data of the request, including the status bar at the bottom: "HTTP Request HTTP-Version (http.request.version), 8 bytes".

Response Version: HTTP/1.1

The image shows a Wireshark packet capture of an HTTP response. The packet list on the left shows a 200 OK response at time 7245.47. The packet details pane on the right shows the response structure:

- Frame 7245: Packet, 241 bytes on wire (1928 bits), 241 bytes captured (1928 bits) on interface DeviceN
- Ethernet II, Src: d2:6c:d5:22:00:3a (d2:6c:d5:22:00:3a), Dst: AzureWaveTec_2e:a8:54 (9c:c7:d3:2e:a8:54)
- Internet Protocol Version 4, Src: 43.205.147.226, Dst: 10.31.226.96
- Transmission Control Protocol, Src Port: 80, Dst Port: 63702, Seq: 20256, Ack: 720, Len: 187
- [13 Reassembled TCP Segments (14582 bytes): #7070(1275), #7071(1275), #7072(379), #7073(1275), #7074(1275)]
- Hypertext Transfer Protocol
 - HTTP/1.1 200 OK
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Date: Sun, 06 Sep 2026 12:20:49 GMT
 - Content-Type: text/css
 - Content-Length: 14143
 - Connection: keep-alive
 - Last-Modified: Thu, 12 May 2022 13:04:34 GMT
 - Accept-Ranges: bytes
 - ETag: "5616f9d3066d810"
 - Server: Microsoft-IIS/10.0
 - Access-Control-Allow-Origin: *
 - Access-Control-Allow-Headers: Content-Type, Accept, x-api-key
 - Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS
 - X-Content-TypeOptions: nosniff

The packet bytes pane on the right shows the raw data of the response, including the status bar at the bottom: "HTTP Response HTTP-Version (http.response.version), 8 bytes".

9. What is the status code and phrase returned from the server to your browser?

→ Status code:200

→ Response phrase:OK

The image shows a Wireshark network traffic capture. The top pane displays a list of network packets. The selected packet is an HTTP response (No. 7245) with a status code of 200 OK. The middle pane shows the details of this packet, including the HTTP response structure and the status code/phrase. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
6584	47.29077100	10.31.226.96	43.205.147.226	HTTP	413	GET /assets/vendor/animate/animate.min.css HTTP/1.1
6764	47.418603700	43.205.147.226	10.31.226.96	HTTP	814	HTTP/1.1 200 OK (text/css)
6768	47.418629200	43.205.147.226	10.31.226.96	HTTP	319	HTTP/1.1 200 OK (text/css)
6803	47.425644200	10.31.226.96	43.205.147.226	HTTP	426	GET /assets/vendor/fancybox/css/jquery.fancybox.min.css HTTP/1.1
6804	47.425661300	10.31.226.96	43.205.147.226	HTTP	426	GET /assets/vendor/owlcarousel/css/owl.carousel.min.css HTTP/1.1
6808	47.426270700	43.205.147.226	10.31.226.96	HTTP	194	HTTP/1.1 200 OK (text/css)
6833	47.432557700	10.31.226.96	43.205.147.226	HTTP	415	GET /assets/vendor/swiper/css/swiper.min.css HTTP/1.1
6996	47.475403100	43.205.147.226	10.31.226.96	HTTP	1299	HTTP/1.1 200 OK (text/css)
7031	47.482586600	10.31.226.96	43.205.147.226	HTTP	411	GET /assets/vendor/swiper/css/swiper.css HTTP/1.1
7088	47.520735900	43.205.147.226	10.31.226.96	HTTP	205	HTTP/1.1 200 OK (text/css)
7115	47.531281700	43.205.147.226	10.31.226.96	HTTP	753	HTTP/1.1 200 OK (text/css)
7143	47.536158300	10.31.226.96	43.205.147.226	HTTP	396	GET /css/lightgallery.css HTTP/1.1
7144	47.536186000	10.31.226.96	43.205.147.226	HTTP	396	GET /assets/css/style.css HTTP/1.1
7225	47.576387500	43.205.147.226	10.31.226.96	HTTP	219	HTTP/1.1 200 OK (text/css)
7242	47.579986500	43.205.147.226	10.31.226.96	HTTP	126	HTTP/1.1 200 OK (text/html)
7245	47.580215800	43.205.147.226	10.31.226.96	HTTP	241	HTTP/1.1 200 OK (text/css)

Frame 7245: Packet, 241 bytes on wire (1928 bits), 241 bytes captured (1928 bits) on interface \Device...

Ethernet II, Src: d2:6c:d5:22:00:3a (d2:6c:d5:22:00:3a), Dst: AzureWaveTec_2e:a8:54 (9c:c7:d3:2e:a8:54)

Internet Protocol Version 4, Src: 43.205.147.226, Dst: 10.31.226.96

Transmission Control Protocol, Src Port: 80, Dst Port: 63702, Seq: 20256, Ack: 720, Len: 187

[13 Reassembled TCP Segments (14582 bytes): #7070(1275), #7071(1275), #7072(370), #7073(1275), #7074(1275)]

Hypertext Transfer Protocol

Response Version: HTTP/1.1

Status Code: 200

[Status Code Description: OK]

Response Phrase: OK

Date: Sun, 06 Sep 2026 12:20:49 GMT\r\n

Content-Type: text/css\r\n

Content-Length: 14143\r\n

Connection: keep-alive\r\n

Last-Modified: Thu, 12 May 2022 13:04:34 GMT\r\n

Accept-Ranges: bytes\r\n

ETag: "5616f9d3066d81:0"\r\n

Server: Microsoft-IIS/10.0\r\n

Access-Control-Allow-Origin: *\r\n

Access-Control-Allow-Headers: Content-Type, Accept, x-api-key\r\n

Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS\r\n

X-Content-TypeOptions: nosniff\r\n

Packet (241 bytes) Reassembled TCP (14582 bytes)

HTTP Response Status Code (http.response.code). 3 bytes

Packets: 15875 - Displayed: 150 (0.9%) - Dropped: 0 (0.0%) Profile: Default

10. What languages does your browser indicate to the server that it can accept? Which header line is used to indicate this information?

➔ Accept-Language: en-US,en;q=0.9,gu;q=0.8\r\n

Accept Language in Header line

The image shows a Wireshark packet capture of an HTTP GET request. The packet list pane shows a packet of 42 bytes. The packet details pane shows the following structure:

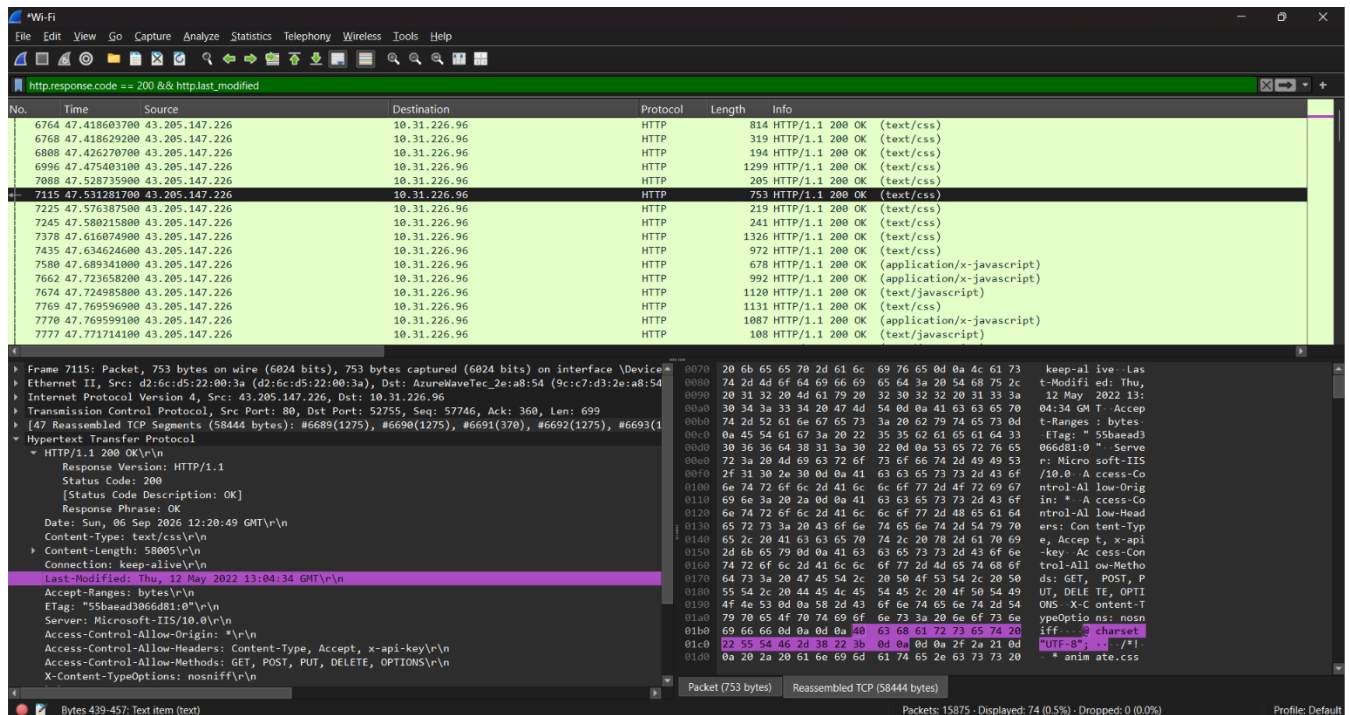
```

Frame 7144: Packet, 396 bytes on wire (3168 bits), 396 bytes captured (3168 bits) on interface \Device\NPF...
Ethernet II, Src: AzureWaveTec_2e:a8:54 (9c:c7:d3:2e:a8:54), Dst: d2:6c:d5:22:00:3a (d2:6c:d5:22:00:3a)
Internet Protocol Version 4, Src: 10.31.226.96, Dst: 43.205.147.226
Transmission Control Protocol, Src Port: 52755, Dst Port: 80, Seq: 360, Ack: 58445, Len: 342
Hypertext Transfer Protocol
  GET /assets/css/style.css HTTP/1.1\r\n
    Request Method: GET
    Request URI: /assets/css/style.css
    Request Version: HTTP/1.1
    Host: www.gtu.ac.in\r\n
    Connection: keep-alive\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/110.0.0.0 Safari/537.36\r\n
    Referer: http://www.gtu.ac.in/\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9,gu;q=0.8\r\n
  \r\n
[Response in frame: 9307]
[Full request URI: http://www.gtu.ac.in/assets/css/style.css]

```

The 'Accept-Language' header is highlighted in the packet details pane, showing the value 'en-US,en;q=0.9,gu;q=0.8\r\n'.

➔ Last-Modified: Thu, 12 May 2022 13:04:34 GMT\r\n



12. How many bytes of content (size of file) are returned to your browser? Which header line is used to indicate this information?

(When the server sends the webpage (HTML file) to your browser, what is the size of that file?)

➔ Content-Length: 18679\r\n

The image shows a Wireshark packet capture of an HTTP response. The packet list on the left shows a packet of 780 bytes. The packet details pane on the right shows the following structure:

- Frame 13929: Packet, 780 bytes on wire (6240 bits), 780 bytes captured (6240 bits) on interface 'Device'
 - Ethernet II, Src: d2:6c:d5:22:00:3a (d2:6c:d5:22:00:3a), Dst: AzureWaveTec_2e:a8:54 (9c:c7:d3:2e:a8:54)
 - Internet Protocol Version 4, Src: 43.205.147.226, Dst: 10.31.226.96
 - Transmission Control Protocol, Src Port: 80, Dst Port: 51144, Seq: 8590888, Ack: 4626, Len: 726
 - [16 Reassembled TCP Segments (19119 bytes): #13914(1275), #13915(1275), #13916(1275), #13917(543), #13918(1275), #13919(1275), #13920(1275), #13921(1275), #13922(1275), #13923(1275), #13924(1275), #13925(1275), #13926(1275), #13927(1275), #13928(1275), #13929(726)]
 - Hypertext Transfer Protocol
 - HTTP/1.1 200 OK\r\n
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Date: Sun, 06 Sep 2026 12:20:51 GMT\r\n
 - Content-Type: image/png\r\n
 - Content-Length: 18679\r\n**
 - Connection: keep-alive\r\n
 - Last-Modified: Thu, 12 May 2022 13:04:30 GMT\r\n
 - Accept-Ranges: bytes\r\n
 - ETag: "61403ad1066d81:0"\r\n
 - Server: Microsoft-IIS/10.0\r\n
 - Access-Control-Allow-Origin: *\r\n
 - Access-Control-Allow-Headers: Content-Type, Accept, x-api-key\r\n
 - Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS\r\n
 - X-Content-TypeOptions: nosniff\r\n

The packet bytes pane on the right shows the raw data of the packet, including the HTTP response structure and the image data.

HTTP Conditional GET/Response Interaction

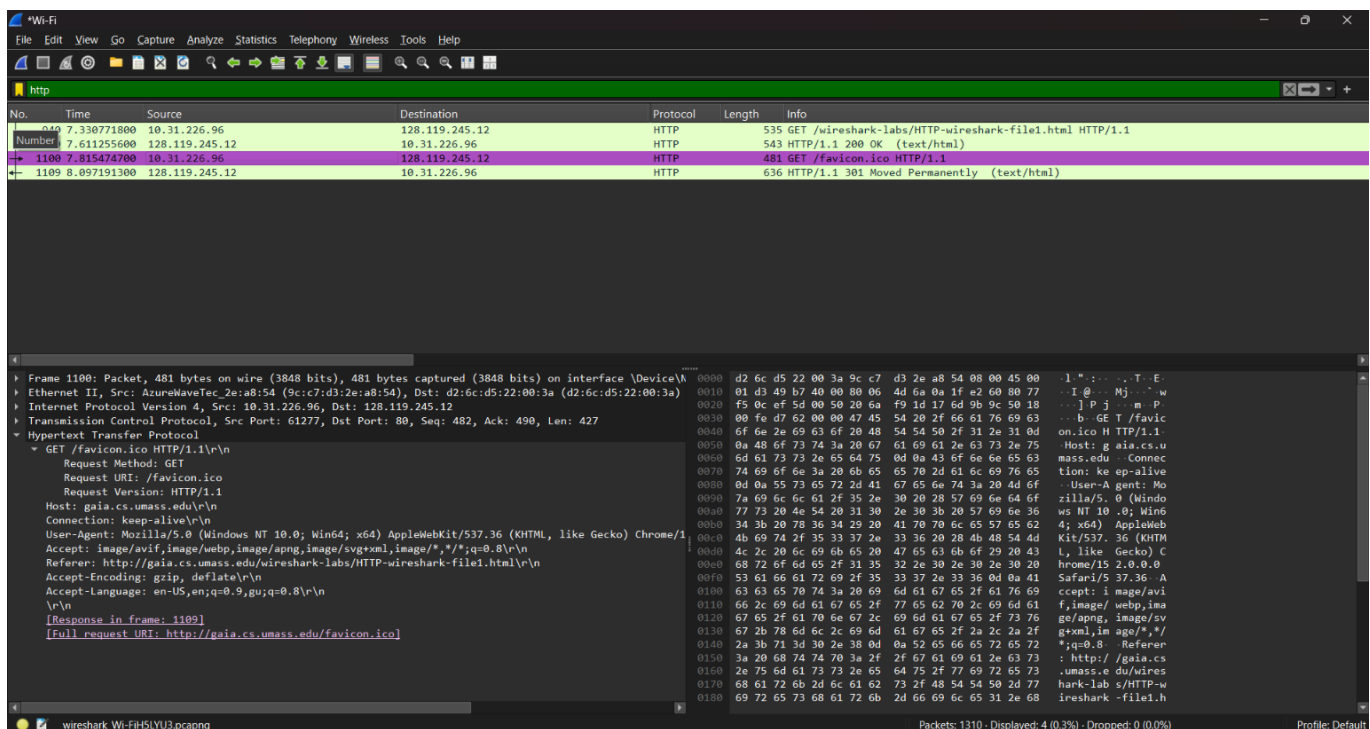
13. Start up the Wireshark packet sniffer, and make sure that “http” is in the displayfilter, so that only captured HTTP messages will be displayed in the packet-list pane.
14. Enter the following URL into your browser:

<http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html>

Your browser should display a very simple HTML file.

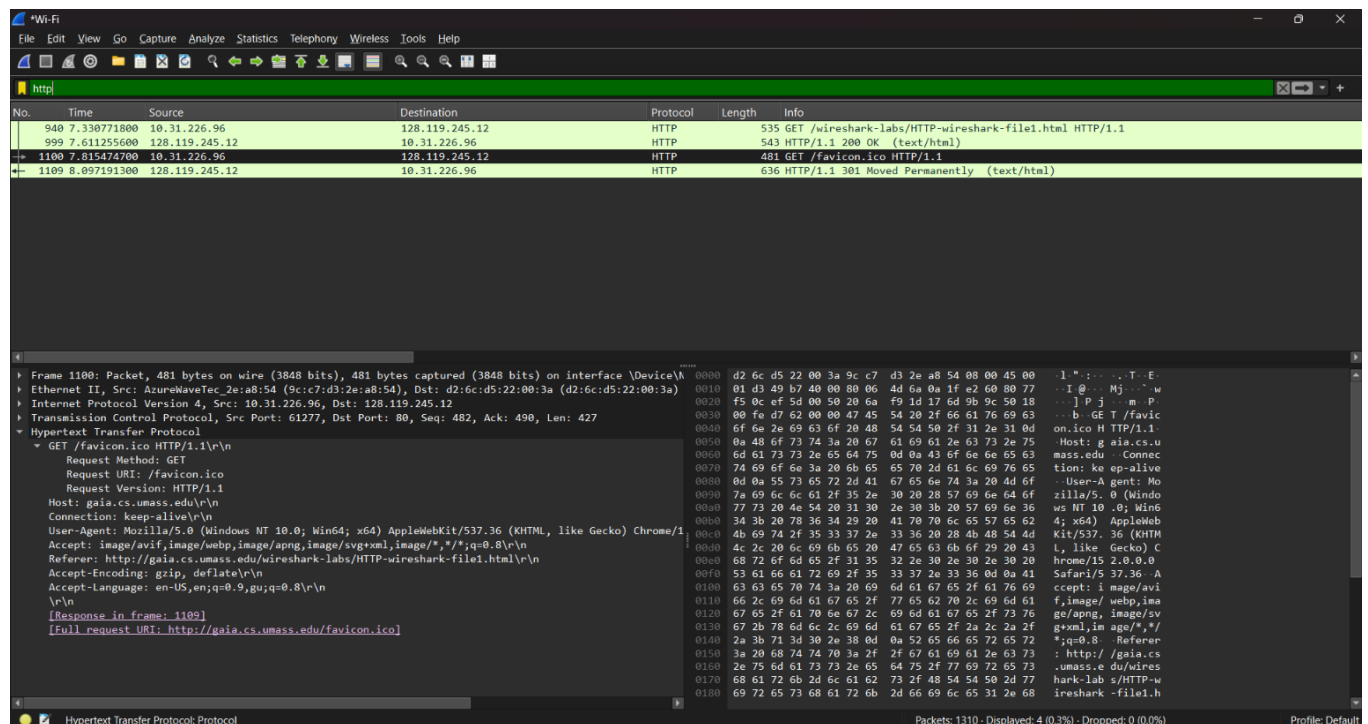
15. Quickly enter the same URL into your browser again (or simply select the refresh button on your browser).

16. Stop Wireshark packet capture.



17. Inspect the contents of the first HTTP GET request from your browser to the server. Is there an “IF-MODIFIED-SINCE” header line in the HTTP GET message? Why or why not? (1st link - no)

Ans: No. If-Modified-Since is not present because the browser is requesting the page for the first time. It has no previous copy of the page, so it simply asks the server to send the page.



18. Inspect the contents of the server response. Has the server explicitly returned the contents of the file? How can you tell?

➔- Yes, the server has returned the contents of the file. Extend line based text data: Congratulation you've download the file

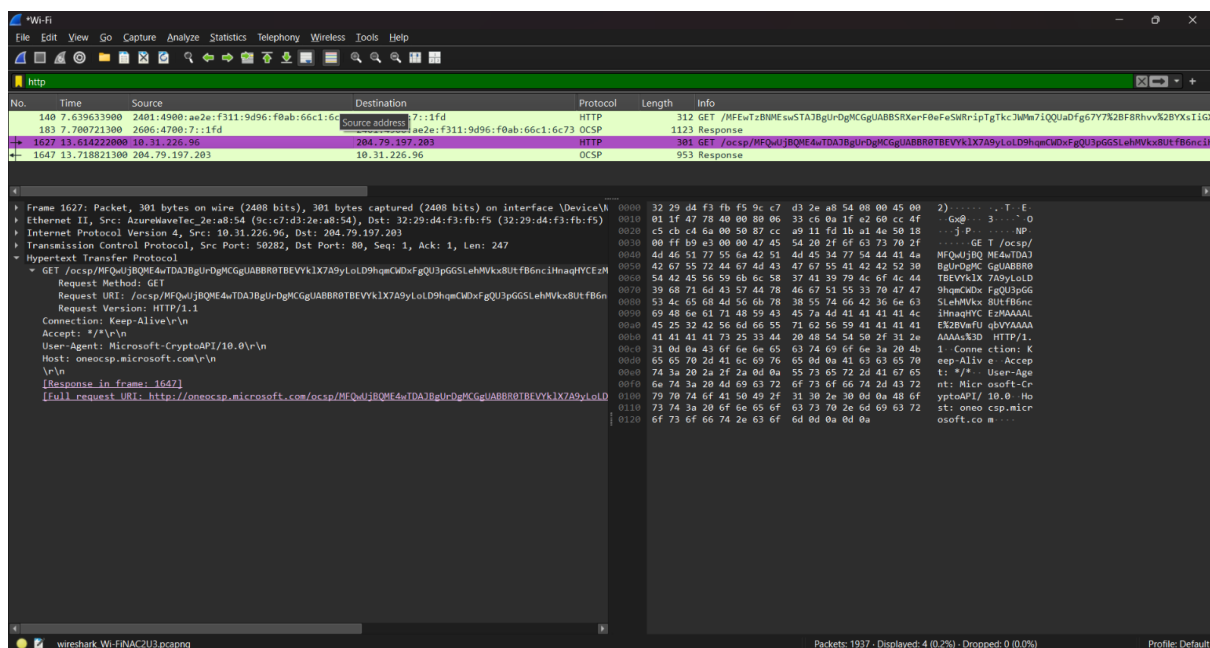
The image shows a Wireshark packet capture of an HTTP response. The packet list at the top shows a GET request (No. 140) and a response (No. 183). The packet details pane for the response (No. 183) shows the following structure:

- Frame 183: Packet, 1123 bytes on wire (8984 bits), 1123 bytes captured (8984 bits) on interface Device
- Ethernet II, Src: 32:29:d4:f3:fb:f5 (32:29:d4:f3:fb:f5), Dst: AzureWaveTec_2a:a8:54 (08:c7:d3:2e:a8:54)
- Internet Protocol Version 6, Src: 2606:4700:7::1fd, Dst: 2401:4900:ae2e:f311:9d96:f0ab:66c1:6c73
- Transmission Control Protocol, Src Port: 80, Dst Port: 56922, Seq: 1, Ack: 239, Len: 1049
- Hypertext Transfer Protocol
 - HTTP/1.1 200 OK\r\n
 - Date: Mon, 07 Sep 2026 03:30:52 GMT\r\n
 - Content-Type: application/octet-stream\r\n
 - Content-Length: 727\r\n
 - Connection: keep-alive\r\n
 - Cache-Control: max-age=7200\r\n
 - Age: 5324\r\n
 - Accept-Ranges: bytes\r\n
 - last-modified: Mon, 07 Sep 2026 02:02:07 GMT\r\n
 - cf-cache-status: HIT\r\n
 - Server: cloudflare\r\n
 - CF-RAY: a372a324cd81efd8-BOM\r\n
 - \r\n
 - [Request in frame: 140]
 - [Time since request: 61.087400 milliseconds]
 - [Request URI: /MFewTzBNMfeswSTA7BgUrDgMcGgUABBSRXerF0eFeSWRripTgTkc7hMm7iQQUaDfg67Y7K2BF8RhhvX2BYXsIIGFu]
 - [Full request URI: http://ocsp.digicert.com/MFewTzBNMfeswSTA7BgUrDgMcGgUABBSRXerF0eFeSWRripTgTkc7hMm7iQQUaDfg67Y7K2BF8RhhvX2BYXsIIGFu]
 - File Data: 727 bytes
 - Online Certificate Status Protocol
 - responseStatus: successful (0)
 - responseBytes
 - Response Type Id: 1.3.6.1.5.7.48.1.1 (id-pkix-ocsp-basic)
 - BasicOCSPResponse

The packet bytes pane shows the raw data of the response, which includes the text "Congratulation you've download the file".

19. Now inspect the contents of the second HTTP GET request from your browser to the server. Is there an “IF-MODIFIED-SINCE:” header line in the HTTP GET message? If so, what information follows the “IF MODIFIED-SINCE:” header line? (second link - yes)

➔ Yes. If-Modified-Since is present because the browser already has a copy of the page. It asks the server whether the page has changed since the last time it was downloaded.



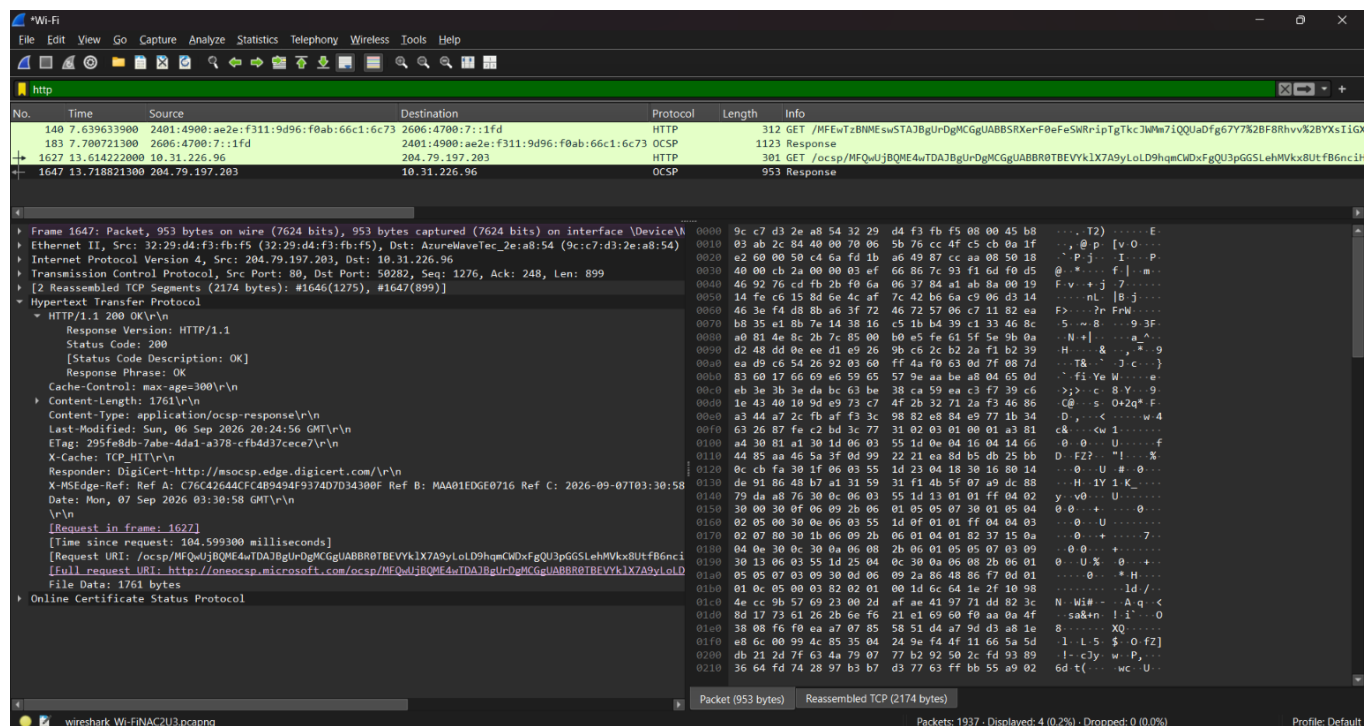
(ans of next que) If it has not changed, the server replies with 304 Not Modified. If yes, the server sends the updated file (200 OK).

20. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Has the server explicitly returned the contents of the file? Explain.

→ Status code:200

→ Response phrase: Not modified

No, the server has not returned the contents of the file as it is the response of the second request sent by the browser.

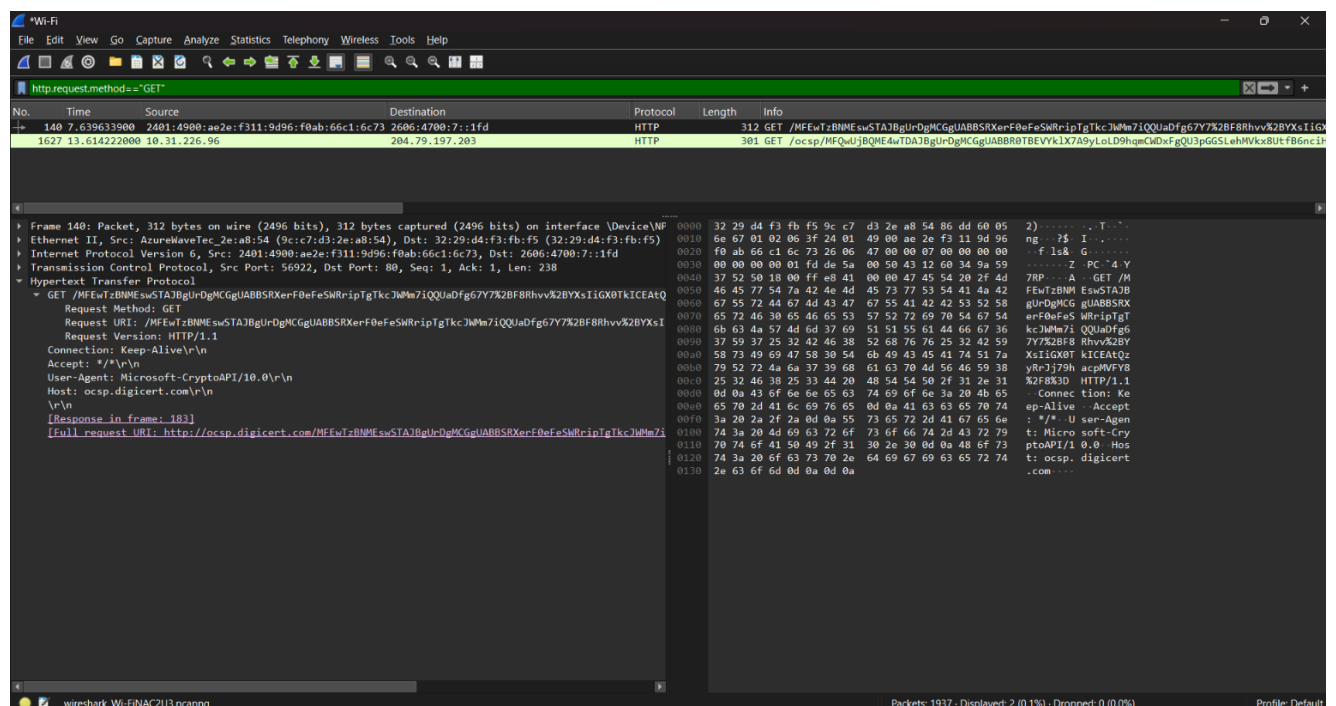


21. How many HTTP GET request messages has your browser sent? Which packet in the trace contains the request for the Bill of Rights?

http.request.method=="GET"

(The packet number of that HTTP GET packet is the answer.)

➔ Total HTTP GET request messages :- 2



22. Which packet in the trace contains the status code and phrase associated with the response to the HTTP GET request? What is the status code and phrase in the response?

Or

(Which packet contains the server's response to the HTTP GET request? What is the status code and status phrase in that response?)

➔ Status Code :- 200 , Response Phrase :- OK

➔ Packet no of server response is : 1937

The image shows a Wireshark packet capture of an HTTP GET request and its response. The packet list on the left shows three packets:

- Packet 140: HTTP GET request from 2401:4900:ae2e:f311:9d96:f0ab:66c1:6c73 to 2606:4700:7::1fd.
- Packet 183: HTTP GET request from 2606:4700:7::1fd to 2401:4900:ae2e:f311:9d96:f0ab:66c1:6c73.
- Packet 1937: HTTP 200 OK response from 2401:4900:ae2e:f311:9d96:f0ab:66c1:6c73 to 2606:4700:7::1fd.

The packet details pane for packet 1937 shows the following information:

- Frame 183: Packet, 1123 bytes on wire (8984 bits), 1123 bytes captured (8984 bits) on interface \Device\NPF{...}
- Ethernet II, Src: AzureWaveTec_2e:a8:54 (0e:c7:d3:2e:a8:54), Dst: 2401:4900:ae2e:f311:9d96:f0ab:66c1:6c73
- Internet Protocol Version 6, Src: 2606:4700:7::1fd, Dst: 2401:4900:ae2e:f311:9d96:f0ab:66c1:6c73
- Transmission Control Protocol, Src Port: 80, Dst Port: 56922, Seq: 1, Ack: 239, Len: 1049
- Hypertext Transfer Protocol
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Date: Mon, 07 Sep 2026 03:30:52 GMT\r\n
 - Content-Type: application/ocsp-response\r\n
 - Content-Length: 727\r\n
 - Connection: keep-alive\r\n
 - Cache-Control: max-age=7200\r\n
 - Age: 5324\r\n
 - Accept-Ranges: bytes\r\n
 - Last-Modified: Mon, 07 Sep 2026 02:02:07 GMT\r\n
 - cf-cache-status: HIT\r\n
 - Server: cloudflare\r\n
 - CF-RAY: a372a324cd81efdb-BOM\r\n
 - [Request in frame: 140]
 - [Time since request: 61.087400 milliseconds]
 - [Request URI: /MFwTzBNMEswSTAjBgUrDgMcGUABBSRXerF0eFeSWRipTgTkcJwMw7lQQUaDfg6Y7YX2BFRhvvX2BYXsIIG]
 - [Full request URI: http://ocsp.digicert.com/MFwTzBNMEswSTAjBgUrDgMcGUABBSRXerF0eFeSWRipTgTkcJwMw7lQQUaDfg6Y7YX2BFRhvvX2BYXsIIG]
 - File Data: 727 bytes

The packet bytes pane shows the raw data of the response, including the status line and headers.

23. Discuss http GET and RESPONSE.

HTTP GET Request:

An HTTP GET request is sent by the browser (client) to the server to request a web page or other resource. It contains information such as the requested URL, HTTP version, and request headers.

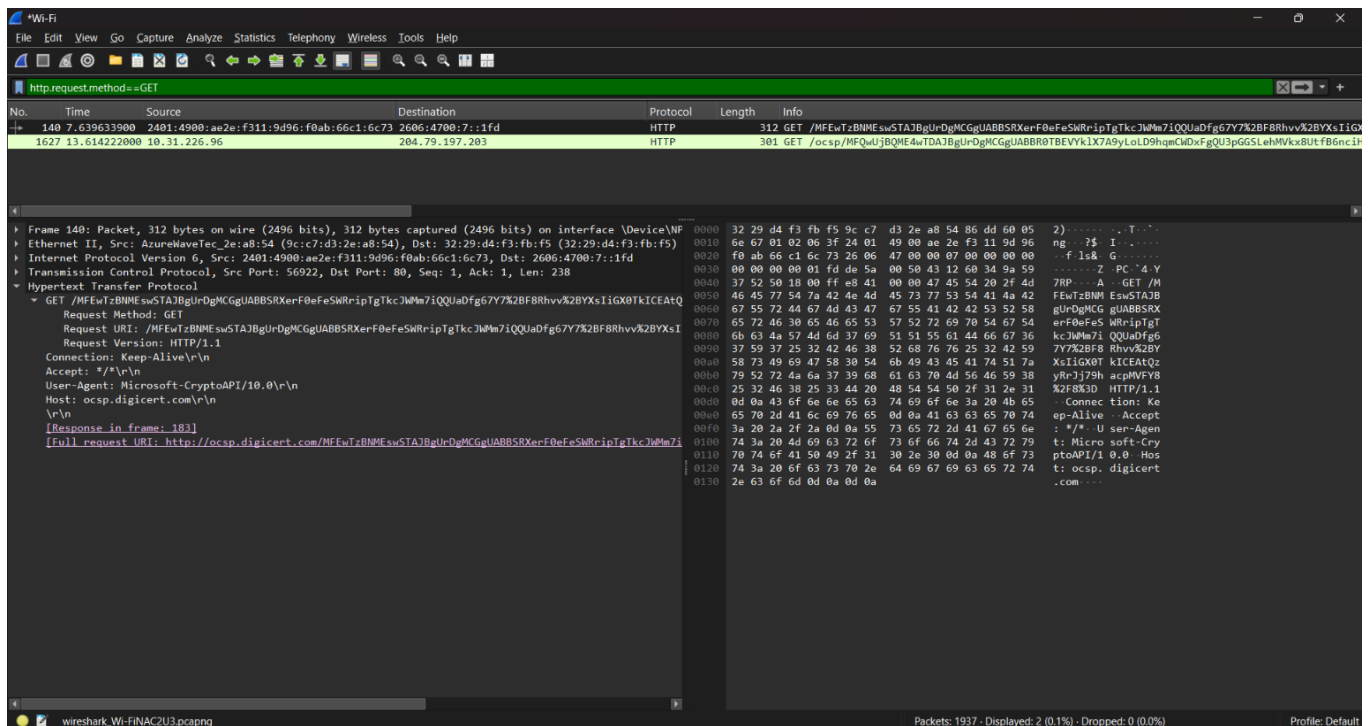
HTTP Response:

An HTTP Response is sent by the server back to the browser after receiving the GET request. It contains the status code (such as 200 OK or 304 Not Modified), response headers, and, if required, the requested web page or file.

24.Count the total number of HTTP GET requests.

➔ Enter in filter: `http.request.method==GET` , then count total -----
marked.

Displayed: 2



25. How much time elapsed between the first HTTP GET request from client and the HTTP response (OK) from server?

➔ `http.request.method==GET || http.response`

(Response Time – GET Time = Elapsed Time)

(How much time did the server take to reply after the browser sent the first HTTP GET request?)

26. Identify the source and destination ports used to transfer the first HTTP GET request

➔ Source Port: 80

➔ Destination Port: 50282

The image shows a Wireshark packet capture of an HTTP GET request. The packet list on the left shows a GET request from 10.31.226.96 to 204.79.197.203 on port 80. The packet details on the right show the destination port as 50282. The packet bytes on the right show the raw data of the request.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
140	7.639633900	2401:4900:ae2e:f311:9d96:f0ab:66c1:6c73	2606:4700:7:1:1fd	HTTP	312	GET /MFwTzBNMEsuSTA3BgUrDgMcGgUABBSRXerF0eFeSWrripTgKcJWw71QQUaDfg67Y7&2BF8Rrhhv&2BYXs1IG
183	7.700721300	2606:4700:7:1:1fd	2401:4900:ae2e:f311:9d96:f0ab:66c1:6c73	OCSP	1123	Response
1627	13.614222000	10.31.226.96	204.79.197.203	HTTP	301	GET /ocsp/MFQwUjBQWE4wTDA3BgUrDgMcGgUABBR0TBVYk1X7A9yLoLD9hqmCwDxFgQU3pGGSLhMVkx8UftfB6nciH
1647	13.718821300	204.79.197.203	10.31.226.96	OCSP	953	Response

Packet Details:

- Frame 1647: Packet, 953 bytes on wire (7624 bits), 953 bytes captured (7624 bits) on interface \Device\NPF{...}
- Ethernet II, Src: 32:29:d4:f3:fb:f5 (32:29:d4:f3:fb:f5), Dst: AzureWaveTec_2e:a8:54 (9c:c7:d3:2e:a8:54)
- Internet Protocol Version 4, Src: 204.79.197.203, Dst: 10.31.226.96
- Transmission Control Protocol, Src Port: 80, Dst Port: 50282, Seq: 1276, Ack: 248, Len: 899
 - Source Port: 80
 - Destination Port: 50282
 - [Stream index: 40]
 - [Stream Packet Number: 7]
 - [Conversation completeness: Incomplete, DATA (15)]
 - [TCP Segment Len: 899]
 - Sequence Number: 1276 (relative sequence number)
 - Sequence Number (raw): 4246447689
 - [Next Sequence Number: 2175 (relative sequence number)]
 - Acknowledgment Number: 248 (relative ack number)
 - Acknowledgment number (raw): 2278337032
 - 0101 ... = Header Length: 20 bytes (5)
 - Flags: 0x018 (PSH, ACK)
 - Window: 16384
 - [Calculated window size: 4194304]
 - [Window size scaling factor: 256]
 - Checksum: 0xc2a [unverified]
 - [Checksum Status: Unverified]
 - Urgent Pointer: 0
 - [Timestamps]
 - [SEQ/ACK analysis]
 - [Client Contiguous Streams: 1]
 - [Server Contiguous Streams: 1]
 - TCP payload (899 bytes)
 - TCP segment data (899 bytes)
 - [2 Reassembled TCP Segments (2174 bytes): #1646(1275), #1647(899)]
- Hypertext Transfer Protocol
- Online Certificate Status Protocol

Packet (953 bytes) Reassembled TCP (2174 bytes)

Destination Port (tcp.dstport), 2 bytes

Packets: 1937 · Displayed: 4 (0.2%) · Dropped: 0 (0.0%) Profile: Default

27. Identify the client and server ip address in the first HTTP GET request?

➔ Apply filter `http.request.method==GET`, see the IP, within this you will find source ip address and destination ip address.

IP Address: 204.79.197.203

Destination Address: 10.31.226.96

The image shows a Wireshark packet capture window. The top pane displays a list of captured packets. The second pane shows the details of the selected packet (No. 1647), which is an HTTP GET request. The third pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
140	7.639633900	2401:4900:ae2e:f311:9d96:f0ab:66c1:6c73	2606:4700:7::1fd	HTTP	312	GET /MFewTzBNMEswSTA7BgUrDgMCGGUABBSRXerF0eFeSWRripTgTkcJWh71QQJaDfg67Y7%2BF8rhvv%2BYXsIIG
183	7.700721300	2606:4700:7::1fd	2401:4900:ae2e:f311:9d96:f0ab:66c1:6c73	OCSP	1123	Response
1627	13.614222000	10.31.226.96	204.79.197.203	HTTP	301	GET /ocsp/MFQwUjBQME4wTDA7BgUrDgMCGGUABBR0TBVYK1X7A9yLoLD9hqmcMDxFGUQ3pG6SLehNVkx8UtfB6nc1h
1647	13.718821300	204.79.197.203	10.31.226.96	OCSP	953	Response

Packet Details (Frame 1647):

- Ethernet II, Src: 32:29:d4:f3:fb:f5 (32:29:d4:f3:fb:f5), Dst: AzureWaveTec_2e:a8:54 (9c:c7:d3:2e:a8:54)
- Internet Protocol Version 4, Src: 204.79.197.203, Dst: 10.31.226.96
- TCP, Src Port: 50282, Dst Port: 80, Seq: 1276, Ack: 248, Len: 899
- Hypertext Transfer Protocol
- Online Certificate Status Protocol

Raw Data (Hex):

```

0010  03 ab 2c 84 40 00 70 06 5b 76 cc 4f c5 cb 0a 1f ...-p- [v-O-
0020  02 60 00 50 c4 6a fd 1b a6 49 87 cc aa 08 50 18 ...P j- I- P-
0030  40 00 cb 2a 00 00 03 ef 66 86 7c 93 f1 6d f0 d5 ...-f- ]- m-
0040  46 92 76 cd fb 2b f0 6a 06 37 84 a1 ab 8a 00 19 ...v-+ j- 7- m-
0050  14 fe c6 15 8d 6e 4c af 7c 42 b6 6a c9 06 d3 14 ...nL- ]B-j-
0060  46 3e f4 d8 8b a6 3f 72 46 72 57 06 c7 11 82 ea ...?n FrW-
0070  b8 35 e1 8b 7e 14 38 16 c5 1b b4 39 c1 33 46 8c ...-B- -9-3F-
0080  a0 81 4e 8c 2b 7c 85 00 b0 a5 fe 61 5f 5e 9b 0a ...M+]- -a-
0090  d2 48 dd 0e ee d1 e9 26 9b c6 2c b2 2a f1 b2 39 ...H- -4- -9-
00a0  ea d9 c6 54 26 92 03 60 ff 4a f0 63 0d 7f 08 7d ...TS- -O- -C-
00b0  83 60 17 66 69 e6 59 65 57 9e aa be a8 04 65 0d ...fi Ye M-
00c0  eb 3e 3b 3e da bc 63 be 38 ca 59 ea c3 f7 39 c6 ...>-c- 8 Y-9-
00d0  1e 43 40 10 9d e9 73 c7 4f 2b 32 71 2a f3 46 86 ...G@-s- 0+2q- F-
00e0  a3 44 a7 2c fb af f3 3c 98 82 e8 84 e9 77 1b 34 ...D- -c- -w-4-
00f0  63 26 87 fe c2 bd 3c 77 31 02 03 01 00 01 23 81 ...- -O- 1- -
0100  a4 30 81 a1 30 1d 06 03 55 1d 0e 04 16 04 16 66 ...- -O- 1- -f
0110  44 85 aa 46 5a 3f 0d 99 22 21 ea 8d b5 db 25 bb ...D- FZ? -" !- -X-
0120  0c fb fa 30 1f 06 03 55 1d 23 04 18 30 16 80 14 ...- -O- -U- #- 0- -
0130  de 91 86 48 b7 a1 31 59 31 f1 4b 5f 07 a9 dc 88 ...-H- 1Y 1 K-
0140  79 da a8 76 30 0c 06 03 55 1d 13 01 01 ff 04 02 ...y- -O- -U- -
0150  30 00 30 0f 06 09 2b 06 01 05 05 07 30 01 05 04 ...- -O- - - -0- -
0160  02 05 00 30 06 06 03 55 1d 0f 01 01 ff 04 04 03 ...- -O- -U- -
0170  02 07 00 30 1b 06 09 2b 06 01 04 01 82 37 15 0a ...- -O- - - -7- -
0180  04 0e 30 0c 30 0a 06 08 2b 06 01 05 05 07 03 09 ...- -O- - - -+ -
0190  30 13 06 03 55 1d 25 04 0c 30 0a 06 08 2b 06 01 ...- -O- -U- - - -+ -
01a0  05 05 07 03 09 30 0d 06 09 2a 86 48 86 f7 0d 01 ...- -O- - - -H- -
01b0  01 0c 05 00 03 02 02 01 00 1d 0c 04 1e 2f 10 98 ...- -O- - - -1d- /-
01c0  4a cc 0b 57 69 23 00 2d af aa 41 97 71 dd 02 3c ...N- Mi# - - -A- q- <
01d0  8d 17 73 61 26 2b 6e f6 21 e1 69 60 f0 aa 0a 4f ...-sa+n- !- j- -O-
01e0  38 08 f6 f0 ea a7 07 85 58 51 d4 a7 9d d3 a8 1e ...8- - - -XQ- - - -
01f0  e8 6c 00 99 4c 85 35 04 24 9e f4 4f 11 66 5a 5d ...1- L- S- $- -0- fZ-
0200  db 21 2d 7f 63 4a 79 07 77 b2 92 50 2c fd 93 89 ...l- -c- j- -w- -P- -
0210  36 64 fd 74 28 97 b3 b7 d3 77 63 ff bb 55 a9 02 ...6d- t- - -wc- -U- -
0220  07 49 77 7d ca 6b 99 fb 78 7b ff d2 82 f8 81 25 ...I- )- k- -x- ( - -X-
  
```

Packet (953 bytes) Reassembled TCP (2174 bytes)

Packets: 1937 - Displayed: 4 (0.2%) - Dropped: 0 (0.0%) Profile: Default

28. Identify the client and server MAC address in the first HTTP GET request?

➔ Apply filter `http.request.method==GET`, see the Ethernet , within this you will find source MAC and destination MAC.

Destination: 9c:c7:d3:2e:a8:54

Source: 32:29:d4:f3:fb:f5

